

How to easily convert between units

1 Background Knowledge Check

1. Recall that any number multiplied by 1 gives you that number back:

$$5 \times 1 = 5$$

$$999 \times 1 = 999$$

$$109583 \times 1 = 109583$$

2. Also recall that any fraction with the same numerator and denominator is equal to one:

$$\frac{2}{2} = 1$$

$$\frac{10402.52}{10402.52} = 1$$

Relating this back to (1.), multiplying by these fractions is the same as multiplying by 1.

3. Moreover, you are able to cancel the numerator and denominator of a fraction using a common factor:

$$\frac{2}{4} = \frac{\cancel{2}^1}{\cancel{4}^2} = \frac{1}{2}$$

In the example above, both 4 and 2 are divisible by 2. So, we divide both numbers by 2 to arrive at a simpler fraction of $\frac{1}{2}$.

4. This also works when multiplying fractions. For any number in the equation, look at the opposite (vertically) side of the fraction for numbers you may be able to cancel with:

$$\frac{10}{9} \times \frac{6}{5}$$

10 is a part of the numerators being multiplied. So, we look at the numbers in the denominators. 9 and 10 are not divisible by the same number; however, 10 and 5 are both divisible by 5, so we cancel them out. 10 becomes 2 ($5 \times 2 = 10$), and 5 becomes 1 ($5 \times 1 = 5$).

$$= \frac{\cancel{10}^2}{9} \times \frac{6}{\cancel{5}^1}$$

$$= \frac{2}{9} \times \frac{6}{1}$$

Similarly, 6 and 5 aren't divisible by the same number, but 6 and 9 are both divisible by 3, so we cancel them out, too. Here, 9 becomes 3 ($3 \times 3 = 9$) and 6 becomes 2 ($3 \times 2 = 6$)

$$= \frac{2}{3} \times \frac{6}{1}$$

$$= \frac{2}{3} \times \frac{2}{1}$$

We can't reduce this any further, so we simply multiply the fractions together.

$$= \frac{4}{3}$$

Important Note *There are many ways to convert units; the method shown in this document works with any units that both start at the same 0 point and scale proportionally (get bigger by the same amount). For example, 0 grams is the same as 0 pounds, and every additional pound is an additional 453.592 grams.*

An example of an exception to this method is degrees Celsius and degrees Fahrenheit. Because 0°C is 32°F , both units do not start at 0 simultaneously.

2 Method of converting units

We are going to use the properties of numbers mentioned in the background section to start solving problems with converting units. To do this, we are going to use the "**Given, Conversion equation and Answer**" method (GCA). Let's start with a simple example:

Question: Convert 2000 grams into kilograms

Solution:

1. **Given:**

We start by writing what we are given. In this case, we are given 2000 grams and want to convert them into kilograms:

$$2000\text{ g} = \underline{\hspace{2cm}}\text{ kg}$$

2. **Conversion equation:**

For our conversion equation, we just need to write an equation that relates our current units (grams) to our desired units (kilograms). We know that 1000 grams is 1 kilogram, therefore our conversion is:

$$1000\text{ g} = 1\text{ kg}$$

or

$$1\text{ kg} = 1000\text{ g}$$

Either one works.

3. Answer:

To start our solution, we write our current units and amount:

$$2000\text{ g}$$

We then write a multiplication symbol followed by a fraction:

$$2000\text{ g} \times \frac{?}{?}$$

Next, we want to fill in our fraction. We first focus on the denominator. We want to make our denominator equal to the part in our conversion equation (step 2) that matches our current units. Since our conversion equation was $1000\text{ g} = 1\text{ kg}$, the part that matches our current units (grams) is simply 1000 g .

$$2000\text{ g} \times \frac{?}{1000\text{ g}}$$

Next, to fill in our numerator, we simply take the other part of our conversion equation (1 kg).

$$2000\text{ g} \times \frac{1\text{ kg}}{1000\text{ g}}$$

We now have our equation set up and ready to solve! This equation works because the fraction we are multiplying by has equal amounts in the numerator and denominator since $1000\text{ g} = 1\text{ kg}$. Such a fraction, as discussed in the background review, is just equal to one. Therefore, multiplying by this fraction does not change the amount our numbers represent, rather it only changes the way it is presented (in kilograms instead of grams).

To begin, we should always focus on units first to ensure we are getting the units we want in our answer.

Similarly to numbers, units in the numerator and denominator may be cancelled out with each other, for example, we are able to cancel the grams in 2000 g and 1000 g with each other:

$$2000\cancel{\text{ g}} \times \frac{1\text{ kg}}{1000\cancel{\text{ g}}}$$

Therefore, we now know our final answer will have the units of kilograms, as that is the only unit left in our equation.

$$2000\cancel{\text{ g}} \times \frac{1\text{ kg}}{1000\cancel{\text{ g}}} = ?\text{ kg}$$

Now we can focus on our numbers, this can be interpreted as 2000 times 1 divided by 1000. Which is simply 2:

$$2000\cancel{\text{ g}} \times \frac{1\text{ kg}}{1000\cancel{\text{ g}}} = 2\text{ kg}$$

Therefore, we know that 2000 grams is equal to 2 kilograms. This method can now be applied to any two units that share a starting 0 point and scale proportionally.

Here is a second quick example:

Question: Convert 290.852 L to mL

Solution:

1. Given:

$$290.852 \text{ L} = \underline{\hspace{2cm}} \text{ mL}$$

2. Conversion equation:

$$1 \text{ L} = 1000 \text{ mL}$$

3. Answer:

$$290.852 \text{ L} \times \frac{1000 \text{ mL}}{1 \text{ L}} = ? ?$$

$$290.852 \cancel{\text{L}} \times \frac{1000 \text{ mL}}{1 \cancel{\text{L}}} = ? \text{ mL}$$

$$290.852 \cancel{\text{L}} \times \frac{1000 \text{ mL}}{1 \cancel{\text{L}}} = 290852 \text{ mL}$$

3 Additional Notes

We are able to perform unit conversions on amounts that are described by multiple units by simply treating each unit separately:

$$1 \frac{\cancel{\text{km}}}{\cancel{\text{K}}} \times \frac{1000 \text{ m}}{1 \cancel{\text{km}}} \times \frac{1 \cancel{\text{K}}}{60 \text{ min}} = 16.\bar{6} \frac{\text{m}}{\text{min}}$$

We are also able to chain unit conversions. Take a look at the following conversion, which converts an hour into seconds by using minutes as a middle-ground and "chains" conversions together:

$$1 \cancel{\text{K}} \times \frac{60 \cancel{\text{min}}}{1 \cancel{\text{K}}} \times \frac{60 \text{ s}}{1 \cancel{\text{min}}} = 3600 \text{ s}$$

We can also combine both of these:

$$1 \frac{\cancel{\text{km}}}{\cancel{\text{K}}} \times \frac{1000 \text{ m}}{1 \cancel{\text{km}}} \times \frac{1 \cancel{\text{K}}}{60 \cancel{\text{min}}} \times \frac{1 \cancel{\text{min}}}{60 \text{ s}} = \frac{1000 \text{ m}}{3600 \text{ s}} = 0.2\bar{7} \frac{\text{m}}{\text{s}}$$

Finally, you may see unit conversions such as cm^3 , which is used to describe volume to be converted into m^3 . To get the conversion equation for such an equation, we can start with the conversion between the base units (m and cm) and then apply the exponential power to both sides of the equal sign:

$$1 \text{ m} = 100 \text{ cm}$$

$$(1 \text{ m})^3 = (100 \text{ cm})^3$$

$$1^3 \text{ m}^3 = 100^3 \text{ cm}^3$$

$$1 \text{ m}^3 = 1,000,000 \text{ cm}^3$$

Now this conversion equation may be used as in the examples above.

Practice Problems

1. Convert the following quantities using the provided conversion equations:

- | | |
|----------------------------|---------------------------------------|
| (a) 2901 g to kg | $1\ kg = 1000\ g$ |
| (b) 82 kg to lb | $1\ kg = 2.20462\ lb$ |
| (c) 2.24 kL to μL | $1\ kL = 10^3\ L, 1\ L = 10^6\ \mu L$ |
| (d) 45 cm^3 to mL | $1\ cm^3 = 1\ mL$ |
| (e) 2 $cups$ to $fl\ oz$ | $1\ cup = 8\ fl\ oz$ |
| (f) 92 km to $miles$ | $1\ mile = 1.60934\ km$ |
| (g) 24.225 m^2 to cm^2 | $1\ m = 100\ cm$ |

2. Convert the following quantities using the provided conversion equations:

- | | |
|--|---|
| (a) 35 $\frac{km}{h}$ to $\frac{m}{s}$ | $1\ km = 1000\ m, 1\ h = 3600\ s$ |
| (b) 45 $\frac{L}{s}$ to $\frac{kL}{h}$ | $1\ kL = 1000\ L, 1\ h = 3600\ s$ |
| (c) 57.21 $\frac{fl\ oz}{L}$ to $\frac{cups}{L}$ | $1\ cup = 8\ fl\ oz$ |
| (d) 193 $\frac{pt}{s}$ to $\frac{gal}{min}$ | $1\ gal = 8\ pt, 1\ min = 60\ s$ |
| (e) 492.24 $km \cdot L$ to $dm \cdot mL$ | $1\ km = 10^4\ dm, 1\ L = 1000\ mL$ |
| (f) 3.5 $\frac{mg}{mL}$ to $\frac{kg}{m^3}$ | $1\ mg = 10^{-6}\ kg, 1\ mL = 10^{-6}\ m^3$ |

3. Challenge yourself:

- | | |
|--|-----------------------------------|
| (a) 4 $cups$ to pt to gal | $1\ pt = 2\ cups, 1\ gal = 8\ pt$ |
| (b) 5.35 g of Ne to mol of Ne | $1\ mol\ Ne = 20.18\ g$ |
| (c) $2.99 \times 10^8 \frac{m}{s}$ to $\frac{km}{h}$ | $1\ km = 1000\ m, 1\ h = 3600\ s$ |
| (d) 292.10230 m to nm | $1\ nm = 10^{-9}\ m$ |

Answers

1. Basic Conversions

- 2.901 kg
- 180.779 lb
- $2.24 \times 10^9\ \mu L$
- 45 mL
- 16 $fl\ oz$
- 57.166 $miles$
- 242250 cm^2

2. Compound Units

- 9.722 $\frac{m}{s}$
- 162 $\frac{kL}{h}$
- 7.15125 $\frac{cups}{L}$
- 1447.5 $\frac{gal}{min}$
- $4.9224 \times 10^9\ dm \cdot mL$
- 3.5 $\frac{kg}{m^3}$

3. Challenge

- 4 $cups = 2\ pt = 0.25\ gal$
- 0.2651 $mol\ Ne$
- $1.0764 \times 10^9 \frac{km}{h}$
- $2.9210230 \times 10^{11}\ nm$